

LAB SHEET 8

Experiment Name: Customer Churn Prediction using Decision Tree Classification Steps

1. Select a suitable dataset such as the Telco Customer Churn Dataset, Bank Customer Churn Dataset, or IBM HR Employee Attrition Dataset.
2. Import the required Python libraries and load the dataset into a Pandas DataFrame.
3. Perform data preprocessing by handling missing values, encoding categorical variables, and selecting relevant features for prediction.
4. Split the dataset into training and testing sets using an appropriate train-test ratio (e.g., 80:20).
5. Train a Decision Tree Classifier using the training dataset and configure suitable parameters such as maximum tree depth and splitting criteria (Gini Index or Entropy)
6. Use the trained model to predict customer churn for the testing dataset.
7. Evaluate the model using a Confusion Matrix, Accuracy, Precision, Recall, and F1- Score.
8. Visualize the Decision Tree to understand the decision-making process and identify the most influential features.
9. Analyze the classification results and determine the key factors contributing to customer churn.
10. Summarize the findings and recommend business strategies for improving customer retention based on the model's predictions.

Solutions ->

```
STEP 1 AND 2: Import Libraries and Load Dataset

[2]
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import (
    confusion_matrix,
    ConfusionMatrixDisplay,
    accuracy_score,
    precision_score,
    recall_score,
    f1_score
)

url = "https://raw.githubusercontent.com/aiplanethub/Datasets/master/WA_Fn-UseC_-Telco-Customer-Churn.csv"
df = pd.read_csv(url)
print("Dataset loaded successfully!")
print(df.head())
print("\nShape:", df.shape)
```

Dataset loaded successfully!

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	
0	7590-WINEG	Female	0	Yes	No	1	No	
1	5575-GWDE	Male	0	No	No	34	Yes	
2	3668-QPYBK	Male	0	No	No	2	Yes	
3	7795-CROCH	Male	0	No	No	45	No	
4	9237-HQITU	Female	0	No	No	2	Yes	

	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	
0	No phone service	DSL	No	...	No	
1	No	DSL	Yes	...	Yes	
2	No	DSL	Yes	...	No	
3	No phone service	DSL	Yes	...	Yes	
4	No	Fiber optic	No	...	No	

	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	
0	No	No	No	Month-to-month	Yes	
1	No	No	No	One year	No	
2	No	No	No	Month-to-month	Yes	
3	Yes	No	No	One year	No	

STEP 3 : Data Preprocessing

```
[3] ✓ Os  print("Missing Values:")
print(df.isnull().sum())
df = df.drop('customerID', axis=1)
df['TotalCharges'] = pd.to_numeric(
    df['TotalCharges'],
    errors='coerce'
)
df['TotalCharges'] = df['TotalCharges'].fillna(
    df['TotalCharges'].median()
)
df['Churn'] = df['Churn'].map({
    'Yes': 1,
    'No': 0
})
df = pd.get_dummies(
    df,
    drop_first=True
)
print("\nProcessed Dataset:")
print(df.head())
```

```
*** Missing Values:
customerID      0
gender          0
SeniorCitizen  0
Partner         0
Dependents      0
tenure          0
PhoneService    0
MultipleLines   0
InternetService 0
OnlineSecurity  0
OnlineBackup    0
DeviceProtection 0
TechSupport     0
StreamingTV     0
StreamingMovies 0
Contract        0
PaperlessBilling 0
PaymentMethod   0
```

```
3          True ...          False
4          False ...          False

StreamingTV_Yes StreamingMovies_No internet service StreamingMovies_Yes \
0          False          False          False          False
1          False          False          False          False
2          False          False          False          False
3          False          False          False          False
4          False          False          False          False

Contract_One year Contract_Two year PaperlessBilling_Yes \
0          False          False          True
1          True          False          False
2          False          False          True
3          True          False          False
```

STEP 4 : Split Dataset into Training and Testing Sets

```
[4] ✓ Os  X = df.drop('Churn', axis=1)
y = df['Churn']
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)
print("Training Data:", X_train.shape)
print("Testing Data:", X_test.shape)
```

```
Training Data: (5634, 30)
Testing Data: (1409, 30)
```

STEP 5 : Train Decision Tree Classifier

```
[5] ✓ Os  model = DecisionTreeClassifier(
    criterion='gini',
    max_depth=5,
    random_state=42
)
model.fit(X_train, y_train)
print("Decision Tree model trained successfully!")
```

```
*** Decision Tree model trained successfully!
```

STEP 6 : Predict Customer Churn

```
[6] ✓ Os  y_pred = model.predict(X_test)
print("Predicted Churn:")
print(y_pred[:20])
```

```
Predicted Churn:
[0 1 0 0 0 0 0 0 0 0 0 0 1 0 0 0 1 0 0]
```



STEP 8 : Visualize the Decision Tree

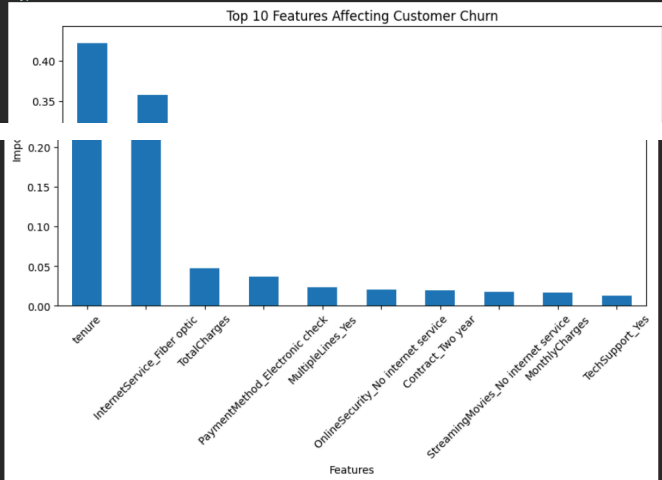
1



STEP 9 : Analyze Important Features

```
[9] importance = pd.Series(
  model.feature_importances_,
  index=X.columns
)
importance = importance.sort_values(
  ascending=False
)
print("Important Features:")
print(importance.head(10))
plt.figure(figsize=(10, 6))
importance.head(10).plot(
  kind='bar'
)
plt.title("Top 10 Features Affecting Customer Churn")
plt.xlabel("Features")
plt.ylabel("Importance")
plt.xticks(rotation=45)
plt.show()
```

```
... Important Features:
tenure                                0.421435
InternetService_Fiber optic          0.357542
TotalCharges                          0.047129
PaymentMethod_Electronic check       0.036761
MultipleLines_Yes                     0.023630
OnlineSecurity_No internet service   0.020669
Contract_Two year                     0.019599
StreamingMovies_No internet service   0.017379
MonthlyCharges                       0.016847
TechSupport_Yes                      0.012988
dtype: float64
```



STEP 10 : Summary and Business Strategies SOLUTION : The Decision Tree model classifies customers into Churn and No Churn categories using customer characteristics. Businesses can use the results to: Identify customers who have a higher predicted risk of churn. Offer discounts or special plans to at-risk customers. Improve customer support for dissatisfied customers. Encourage customers to choose longer-term contracts. Provide personalized offers based on customer behavior. Develop customer-retention campaigns.